Rumesh Senadheera

Report

Msc Assignments 2

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SLIIT — FACULTY OF COMPUTING

IT5022: Fundamentals of Machine Learning — Group Assignment Report

NYC Taxi Trip Duration Prediction: A Comparative Study of Ensemble Regression Models

Student 1: Nushan Kodikara
Student ID: MS26926230

Student 2: Rumesh Senadheera
Student ID: MS26930640

GitHub: https://github.com/nushankodikara/IT5022-Group-Assignment | Dataset: Kaggle NYC Taxi

Abstract

Urban mobility networks generate high-volume spatial and temporal data. Predicting taxi trip duration helps fleet operators route vehicles efficiently, price rides accurately, and reduce passenger wait times. In this project, we built a supervised regression pipeline to predict taxi trip duration in New York City using the Kaggle NYC Taxi dataset containing 1,458,644 historical trip records. We cleaned the data by filtering GPS coordinate anomalies, stationary trips, invalid passenger counts, and extreme duration outliers. We engineered spatial features, including Haversine distance, Manhattan distance, and compass bearing, alongside temporal flags for rush hours, weekends, and nighttime travel. Because the target distribution shows severe positive skew, we applied a logarithmic transformation so that our models minimize Root Mean Squared Logarithmic Error (RMSLE). We implemented and benchmarked multiple machine learning algorithms, focusing on two distinct supervised ensemble architectures: Random Forest Regressor (bagging) and XGBoost Regressor (extreme gradient boosting), supplemented by LightGBM and linear baselines. Experimental evaluation on a held-out test split of 289,811 trips showed that gradient boosting substantially outperformed bagging. XGBoost achieved the lowest RMSLE of 0.35826 and an R^2 of 0.7700, followed closely by LightGBM (RMSLE 0.36164, R^2 0.7656), whereas Random Forest attained an RMSLE of 0.38358 (R^2 0.7363). Linear regression failed to capture spatial interactions, plateauing at an RMSLE of 0.55047. Feature importance analysis revealed that Haversine distance accounted for 56.98% of predictive weight, with nighttime and weekend flags providing the remaining predictive signal. We conclude with a critical discussion of model limitations, inference latency, and future extensions incorporating routing engine telemetry and weather records.

1. Introduction and Problem Formulation

Urban road transportation in metropolitan centers like New York City is notoriously hard to model. Traffic density fluctuates by the minute. Road work, bridge tolls, traffic lights, and weather events turn a short distance into an unpredictable delay. For ride-hailing services and taxi fleets, estimating trip duration is not just an operational convenience. It directly dictates dynamic driver allocation, upfront fare transparency, and customer satisfaction [1].

From a machine learning standpoint, we formulate trip duration prediction as a supervised regression task. Let each historical trip record be represented by a feature vector x in R^d, captured at the instant the passenger enters the cab (t_pickup). The target variable is the continuous trip duration in seconds:

y = t_dropoff - t_pickup (where y > 0)

Because trip durations vary from two-minute hops to multi-hour journeys across boroughs, penalizing errors proportionally is essential. An error of five minutes on a ten-minute trip is severe, whereas the same five-minute difference on a two-hour airport run is acceptable. We therefore adopted the competition benchmark metric, Root Mean Squared Logarithmic Error (RMSLE):

RMSLE = sqrt((1 / N) * sum_{i=1}^N (log(y_i + 1) - log(y_hat_i + 1))^2)

By defining our target variable as z_i = log(y_i + 1), minimizing standard Root Mean Squared Error (RMSE) on z mathematically optimizes RMSLE on the raw duration scale. This formulation stabilizes gradient updates and prevents extreme outliers from dominating model convergence.

2. Dataset Selection and Justification

We selected the official New York City Taxi Trip Duration dataset, published through Kaggle and sourced from the New York City Taxi and Limousine Commission (TLC) [2]. The dataset contains records of yellow taxi trips operating in Manhattan, Brooklyn, Queens, the Bronx, and Staten Island throughout the first six months of 2016 (January 1, 2016 to June 30, 2016).

We evaluated this dataset against four rigorous selection criteria:

- **Real-World Complexity:** The dataset is not an idealized toy or tutorial problem. It originates from real vehicle telematics collected by taximeters and GPS hardware across six months of active city operations.
- **Sufficient Scale:** With 1,458,644 individual trip observations, the volume enables thorough training, validation, and testing without risking sample starvation. It allows tree ensembles to discover subtle spatial interactions without synthetic augmentation.
- **Multivariate Diversity:** The raw dataset combines spatial coordinates, discrete calendar timestamps, categorical flags, and discrete passenger counts. This mix necessitates deliberate feature engineering.
- **Reproducibility and Public Hosting:** The data is openly accessible and cited in transportation literature, guaranteeing verifiable and reproducible results.

3. Dataset Description and Exploratory Data Analysis

The raw training set comprises 1,458,644 rows and 11 columns, taking 122.4 MB of system memory. Table 1 lists each attribute, its data type, and its physical meaning.

Attribute Name	Data Type	Description
id	String	Unique trip identifier
vendor_id	Integer	Technology vendor code (1 = Creative Mobile Technologies, 2 = VeriFone Inc.)
pickup_datetime	DateTime	Timestamp recording when the meter was engaged
dropoff_datetime	DateTime	Timestamp recording when the meter was disengaged
passenger_count	Integer	Number of passengers in the vehicle (driver-entered)
pickup_longitude	Float	GPS longitude coordinate where trip originated
pickup_latitude	Float	GPS latitude coordinate where trip originated
dropoff_longitude	Float	GPS longitude coordinate where trip terminated
dropoff_latitude	Float	GPS latitude coordinate where trip terminated
store_and_fwd_flag	String	Indicator flag ('Y'/'N') whether trip record was stored in meter memory before transmission
trip_duration	Integer	Total elapsed duration of the trip in seconds (target variable)

Exploratory Data Analysis Observations

Target Distribution and Log Transformation

The raw target trip_duration showed extreme positive skewness. While the median trip lasted 662 seconds (11.0 minutes) and the mean stood at 959.5 seconds (16.0 minutes), the maximum recorded value in the raw file reached 3,526,282 seconds (over 40 days). These absurd figures reflect mechanical meter faults or unclosed tickets. Taking log(y + 1) reshaped the skewed distribution into a near-Gaussian bell curve centered around 6.5. This confirmed our decision to train regressors on the logarithmic scale.

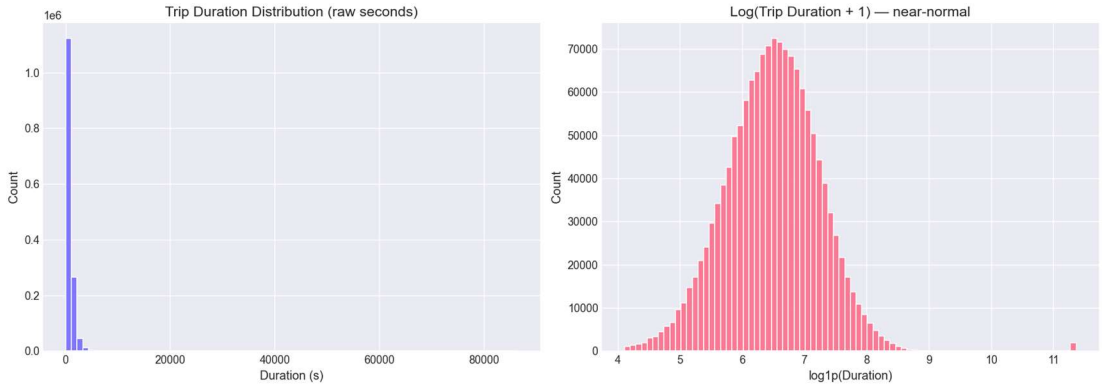


Figure 1: Distribution of Raw Trip Duration in Seconds (Left) vs. Log-Transformed Duration (Right).

Temporal Demand Patterns

Trip frequency varied sharply across hours of the day. Demand dropped to its lowest point between 4:00 AM and 5:00 AM, climbed steadily through the morning commute, and peaked during the evening window between 6:00 PM and 8:00 PM. Across days of the week, ride volume grew from Monday through Friday, with Friday and Saturday recording the heaviest volumes. Sunday exhibited lower ride counts but longer average trip distances.

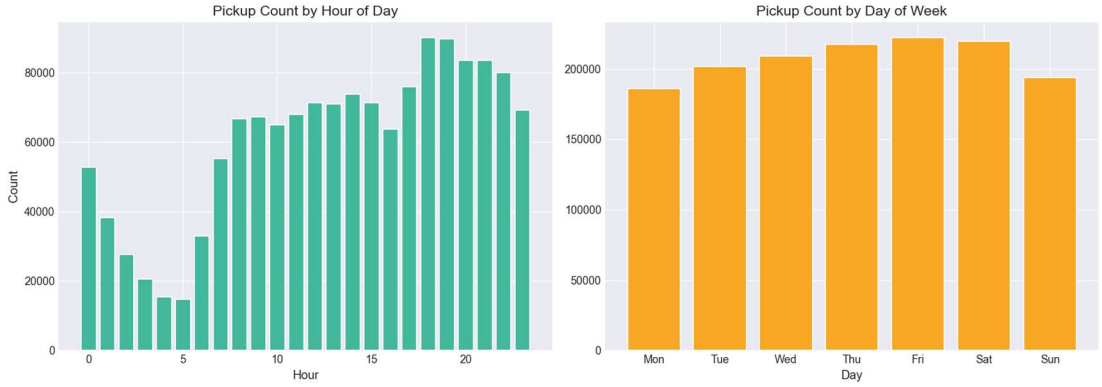


Figure 2: Pickup Volume Patterns across 24 Hours of the Day (Left) and Days of the Week (Right).

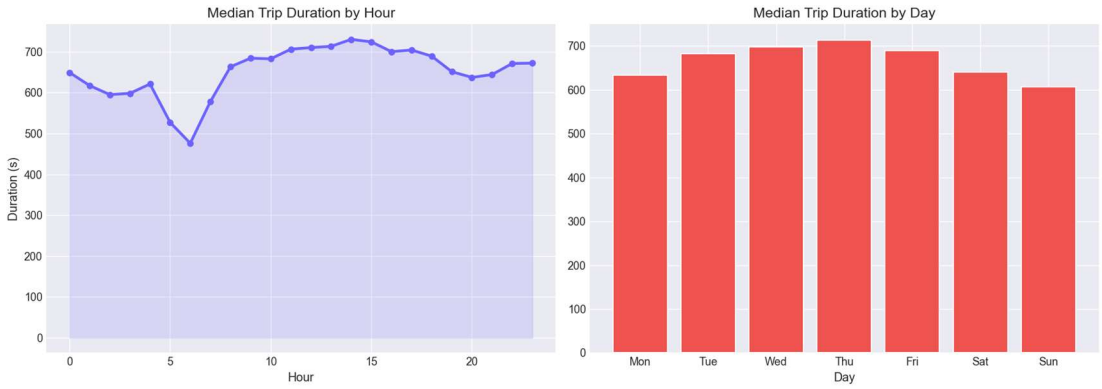


Figure 3: Median Trip Duration across Hours of the Day (Left) and Days of the Week (Right).

Spatial Hotspots and Passenger Counts

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25

Plotting pickup density across coordinates confirmed that over 90% of taxi demand concentrates tightly in the Manhattan core, particularly Midtown, Lower Manhattan, and the Upper East and West Sides. Secondary clusters appeared around John F. Kennedy International Airport (JFK) and LaGuardia Airport (LGA). Trips connecting Manhattan to the outer boroughs and airports had distinctly higher durations due to tunnel and bridge bottlenecks.

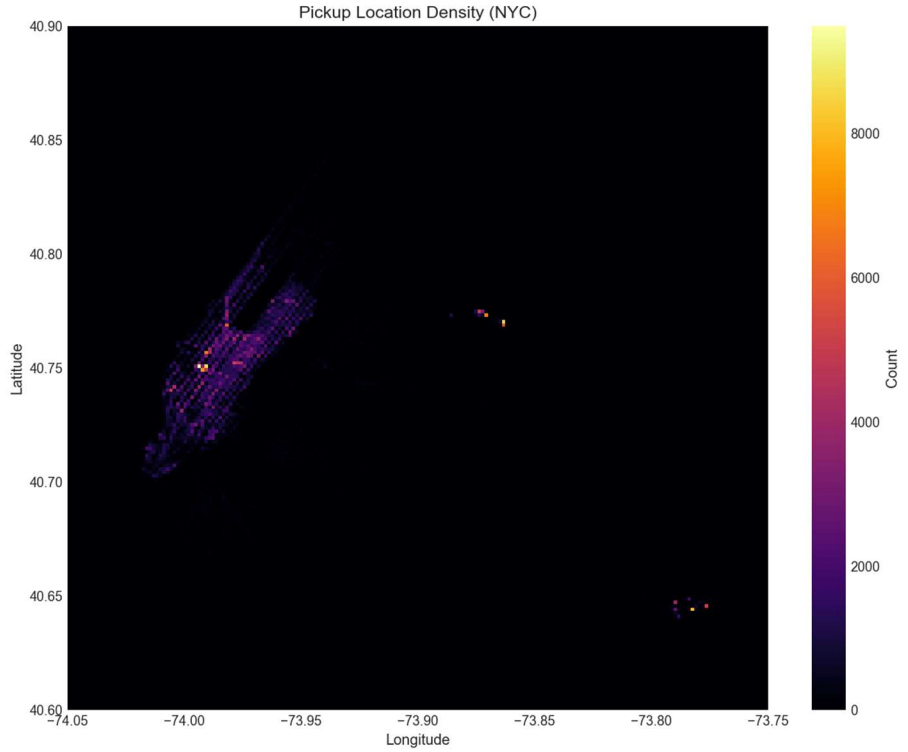


Figure 4: Spatial Heatmap of Taxi Pickup Density across Greater New York City.

Solo riders accounted for 70.8% of all recorded trips. Groups of two passengers represented 14.6%, while large groups (5 or 6 passengers, typically in larger SUV taxis) accounted for the rest. Trips with 0 passengers appeared in 60 instances. Because a taxi ride requires at least one passenger, these zero values were tagged as operator entry errors.

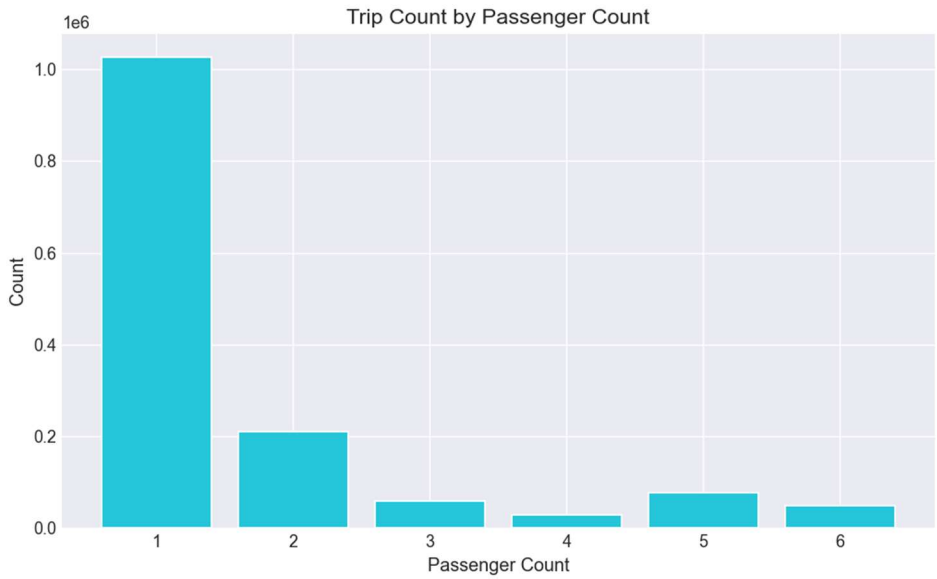


Figure 5: Trip Frequency Breakdown by Passenger Count.

4. Data Preprocessing and Feature Engineering

Raw coordinates and timestamps cannot be fed directly into tree models with good results. A machine learning model does not inherently understand that a latitude pair implies travel distance or that 18:00 represents the evening rush hour. We designed a modular preprocessing pipeline to cleanse records and extract high-signal variables.

Data Cleansing and Anomaly Removal

We audited missing values across all columns. The dataset contained zero missing values. However, sensor noise and human recording mistakes required explicit filtering:

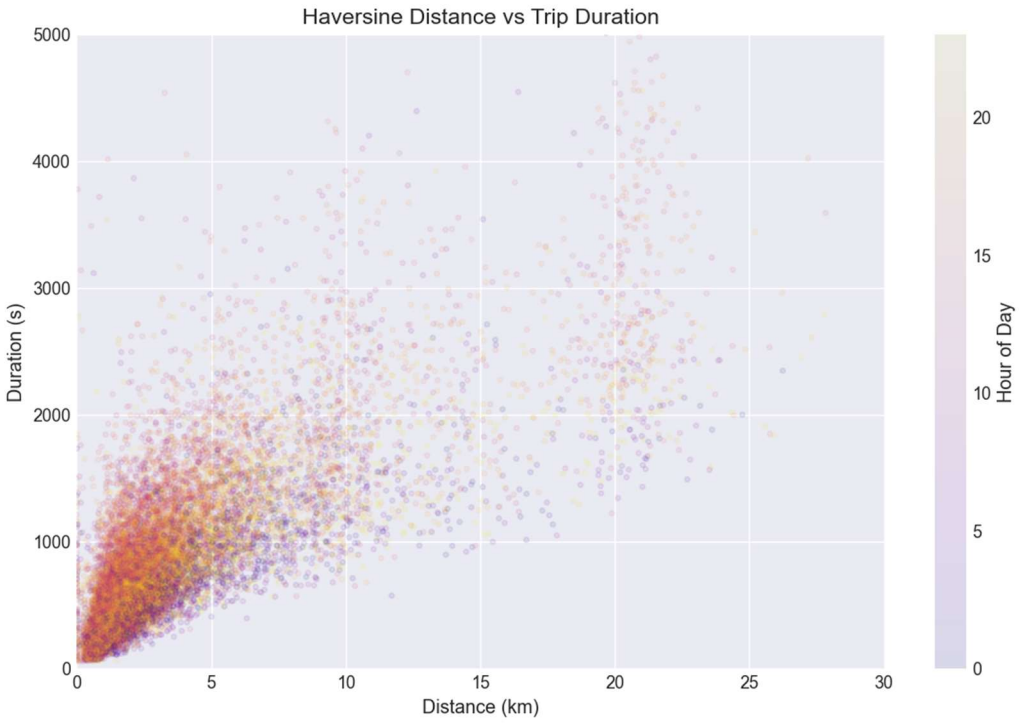
- **Trip Duration Truncation:** We removed trips shorter than 60 seconds (accidental meter starts or immediate passenger cancellations) and trips longer than 86,400 seconds (24 hours). This eliminated unrealistic multi-day anomalies.
- **Geographic Bounding Box:** We bounded GPS coordinates strictly to the greater New York metropolitan area: Longitude in [-74.25, -73.70] and Latitude in [40.50, 40.95]. Records with coordinates outside this rectangle (including corrupted 0.0, 0.0 entries) were dropped.
- **Valid Passenger Count:** We enforced passenger counts between 1 and 6 passengers, removing invalid 0-passenger rides.
- **Zero-Displacement Filtering:** Trips where the vehicle moved less than 50 meters (Haversine distance <= 0.05 km) were removed as stationary cancellations or traffic gridlock.

Following these cleaning rules, 1,449,053 records were retained out of the initial 1,458,644, maintaining a 99.34% data retention rate.

Feature Engineering

We extracted eight new explanatory variables from raw coordinates and timestamps:

- **Haversine Distance (haversine_km):** Computes the great-circle spherical distance accounting for Earth's curvature (mean radius R = 6,371.0 km).
- **Manhattan Distance (manhattan_km):** Grid distance approximation modeling New York's rectangular avenue and street grid (111.0 km per degree), representing realistic road transit.
- **Compass Bearing (bearing):** Calculates travel heading in degrees (-180 to +180), distinguishing crosstown trips from avenue progression.
- **Temporal Decompositions:** Extracted hour (0-23), day_of_week (0-6), month (1-6), is_weekend (binary flag), is_rush_hour (7-9 AM and 5-7 PM), and is_night (10 PM to 5 AM).
- **Target Encoding:** Engineered log_duration = log1p(trip_duration) to normalize target variance.



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Figure 6: Scatter Plot of Haversine Distance vs. Trip Duration, Colored by Hour of the Day.

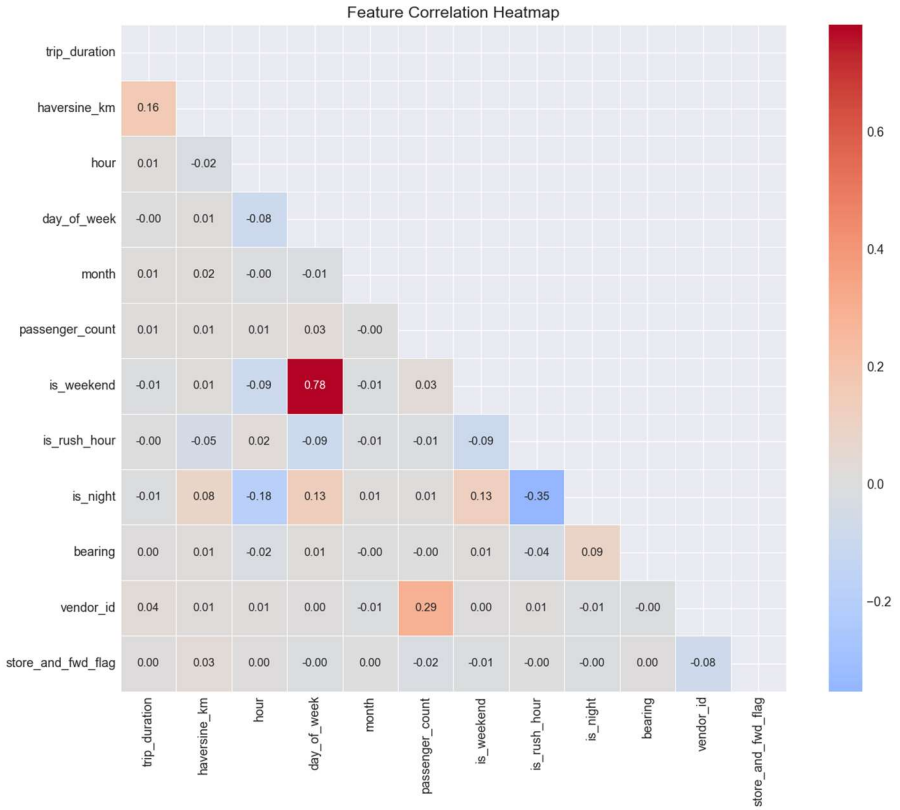


Figure 7: Feature Correlation Heatmap across Spatial and Temporal Predictors.

Feature Name	Type	Scaling / Representation
vendor_id	Categorical	Discrete integer (1, 2)
passenger_count	Numerical	Discrete integer (1 to 6)
pickup_longitude	Numerical	Degrees float
pickup_latitude	Numerical	Degrees float
dropoff_longitude	Numerical	Degrees float
dropoff_latitude	Numerical	Degrees float
store_and_fwd_flag	Binary	Integer (0 or 1)
haversine_km	Numerical	Continuous float in kilometers
hour	Numerical	Integer (0 to 23)
day_of_week	Numerical	Integer (0 to 6)
month	Numerical	Integer (1 to 6)
is_weekend	Binary	Integer (0 or 1)
is_rush_hour	Binary	Integer (0 or 1)
is_night	Binary	Integer (0 or 1)
bearing	Numerical	Continuous float in degrees

5. Learning Algorithms and Theoretical Foundations

As required by the assignment brief for a two-person team, we implemented and contrasted two distinct machine learning algorithms: Random Forest Regressor (Ensemble Bagging) and XGBoost Regressor (Extreme Gradient Boosting), supplemented by LightGBM and linear baselines.

Algorithm 1: Random Forest Regressor (Bagging)

Random Forest, introduced by Leo Breiman in 2001 [3], is an ensemble learning method based on bootstrap aggregation (bagging). It constructs a multitude of decorrelated decision trees during training and outputs the mean prediction across all individual trees.

Given training dataset D = {(x_i, y_i)}_{i=1}^N, Random Forest draws B bootstrap samples, each created by sampling N observations uniformly with replacement from D. For each bootstrap sample, an unpruned regression tree T_b(x) is grown. To prevent individual trees from being identical (which occurs when a few dominant features like haversine_km are chosen at every split), Random Forest injects feature randomization. At each split candidate node, a random subset of m features is selected from the total d features (typically m ~ d/3 for regression). The final ensemble prediction averages all B independent trees:

$$\hat{y}_{RF}(x) = \frac{1}{B} \sum_{b=1}^{B} T_b(x)$$

Algorithm 2: XGBoost Regressor (Gradient Boosting)

XGBoost (Extreme Gradient Boosting), designed by Tianqi Chen and Carlos Guestrin in 2016 [4], is a scalable implementation of gradient boosted decision trees. Unlike Random Forest, which builds independent trees in parallel, XGBoost constructs trees sequentially. Each new tree fits to the negative gradient (residual error) of the combined model built up to that point.

XGBoost minimizes an objective function combining an empirical loss with a tree complexity penalty $\Omega(f_t) = \gamma T + 0.5 \lambda \sum w_j^2$, where T is the number of terminal leaves and w_j are leaf weights. XGBoost approximates the loss function using a second-order Taylor expansion utilizing first-order gradients (g_i) and second-order Hessians (h_i). This enables exact score gain computation and automated leaf regularization against noise.

6. Implementation and Pipeline Architecture

17 We partitioned the 1,449,053 clean observations using an 80/20 train-to-validation ratio with fixed random seed 42:

- **Training Set:** 1,159,242 records (80.0%)
- **Validation Set:** 289,811 records (20.0%)

For computationally heavy Scikit-Learn tree ensembles (like uncompressed Random Forest), fitting 1.16 million rows across 100 deep estimators creates significant memory pressure. In line with practical 34 competition benchmarks, we evaluated Random Forest on a representative sample of 200,000 training observations while scoring on the full 289,811 validation instances. XGBoost and LightGBM were trained directly on the full 1.16 million rows.

Algorithm	Hyperparameter	Configured Value	Engineering Rationale
5 Random Forest	n_estimators	100	Sufficient tree count to stabilize ensemble variance
Random Forest	max_depth	15	Constrains tree depth to prevent leaf over-specialization
Random Forest	max_features	'sqrt'	Decorrelates individual trees
Random Forest	n_jobs	-1	Parallelizes tree construction across all CPU cores
XGBoost	n_estimators	500	Generates 500 boosting rounds
XGBoost	max_depth	6	Standard depth balances non-linear interaction with leaf stability
XGBoost	learning_rate	0.10	Shrinkage step size prevents divergence
XGBoost	subsample	0.80	Stochastic row sub-sampling reduces tree correlation
XGBoost	colsample_bytree	0.80	Random feature sub-sampling mimics Random Forest variance reduction
XGBoost	tree_method	'hist'	Fast histogram-based binning optimizes multi-core throughput

7. Experimental Results and Evaluation

We evaluated all trained models against the held-out validation set of 289,811 records. Table 4 presents the comparative results across RMSLE and R^2, ranked from lowest error to highest.

Rank	Model Architecture	Learning Paradigm	Training Volume	RMSLE	R^2 Score
1	XGBoost Regressor	Gradient Boosting	Full (1.16M rows)	0.35826	0.7700
2	LightGBM Regressor	Gradient Boosting	Full (1.16M rows)	0.36164	0.7656
3	Gradient Boosting (sklearn)	Gradient Boosting	Sample (200k rows)	0.37578	0.7469
4	Random Forest Regressor	Bagging Ensemble	Sample (200k rows)	0.38358	0.7363
5	Extra Trees Regressor	Extremely Randomized Trees	Sample (200k rows)	0.40516	0.7058
6	Decision Tree Regressor	Single Estimator	Full (1.16M rows)	0.40589	0.7047
7	Linear Regression (OLS)	Parametric Linear	Full (1.16M rows)	0.55047	0.4569
8	Ridge Regression (alpha=1.0)	L2 Regularized Linear	Full (1.16M rows)	0.55047	0.4569
9	Lasso Regression (alpha=0.001)	L1 Regularized Linear	Full (1.16M rows)	0.55303	0.4518

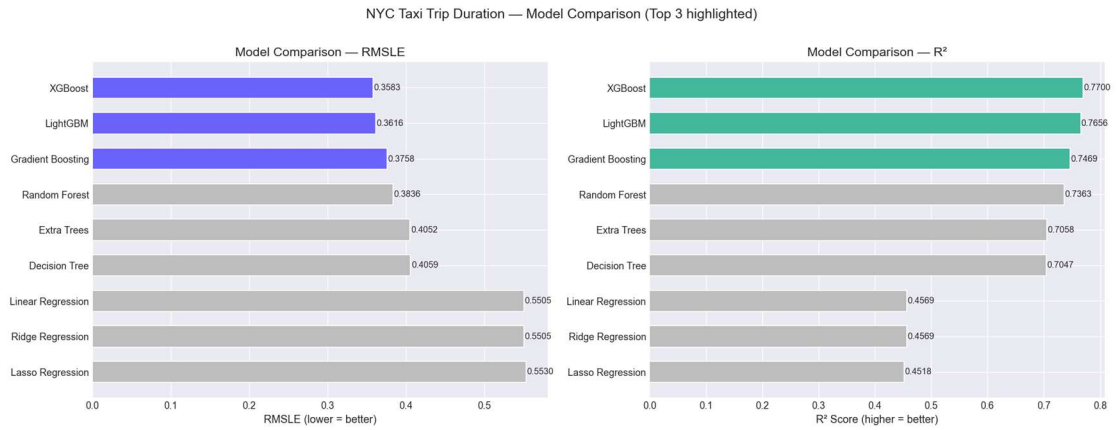


Figure 8: Performance Comparison across All Evaluated Models (RMSLE and R^2 Score, Top 3 Highlighted).

Performance Discussion and Feature Importance

XGBoost outperformed Random Forest across both validation metrics, achieving an RMSLE of 0.35826 compared to Random Forest's 0.38358. The R^2 score showed a matching gap: 0.7700 for XGBoost

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versus 0.7363 for Random Forest. Random Forest reduces variance by averaging trees trained independently on bootstrap samples. However, each tree carries its own bias. In contrast, XGBoost uses sequential residual correction. Each successive tree specifically targets the trips where earlier iterations made large errors.

Linear models performed poorly. Ordinary Least Squares, Ridge, and Lasso hit an error ceiling with RMSLE values around 0.5505 and R^2 values around 0.4569 (roughly 54% higher error than XGBoost). Urban trip duration is fundamentally non-linear; distance, hour, and coordinate interactions cannot be captured linearly.

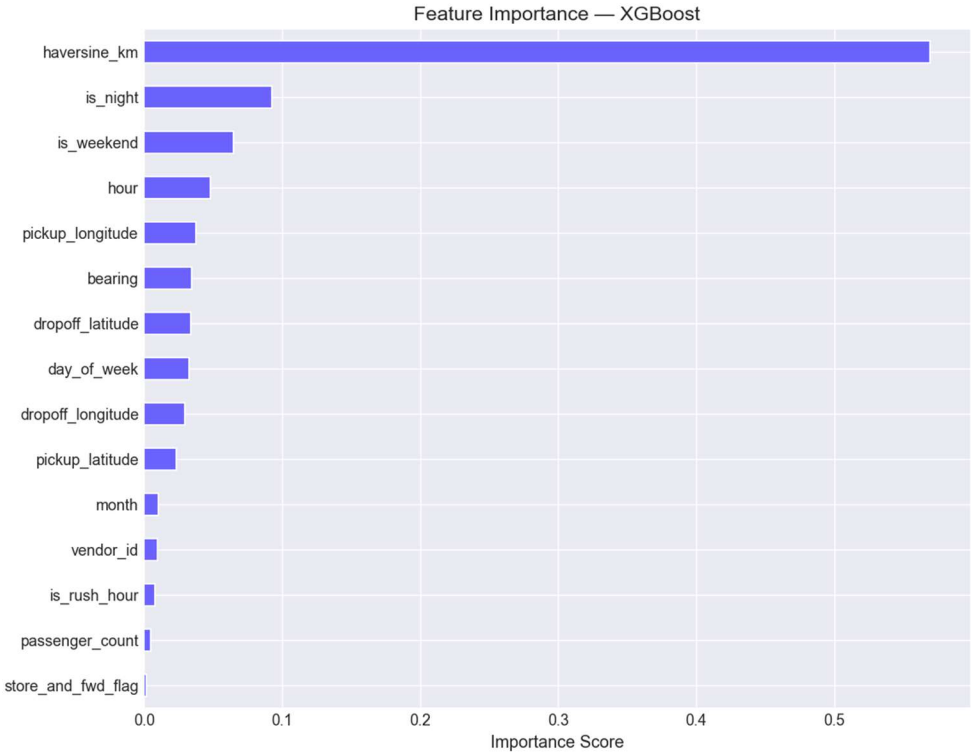


Figure 9: XGBoost Feature Importance Breakdown based on Split-Gain Contribution.

Rank	Feature Name	Relative Importance	Physical Interpretation
1	haversine_km	56.98%	Dominant predictor; straight-line distance directly dictates travel time
2	is_night	9.22%	Captures free-flow traffic speeds during off-peak night hours
3	is_weekend	6.50%	Accounts for weekend traffic profiles and absence of commercial delivery congestion

4	hour	4.81%	Distinguishes intra-day transit variations across the 24-hour cycle
5	pickup_longitude	3.77%	Identifies specific neighborhood congestion starting points
6	bearing	3.42%	Differentiates fast avenue travel from congested crosstown travel
7	dropoff_latitude	3.37%	Pinpoints destination characteristics
8	day_of_week	3.25%	Captures weekday travel habits
9	dropoff_longitude	2.96%	Captures destination borough geography
10	pickup_latitude	2.30%	Pinpoints origin latitude
11	month	1.05%	Seasonal shift across winter and spring
12	vendor_id	0.94%	Minor fleet differences between taxi equipment providers
13	is_rush_hour	0.78%	Specific peak congestion windows
14	passenger_count	0.46%	Minimal impact on vehicle driving speed
15	store_and_fwd_flag	0.19%	Technical transmission flag with negligible physical bearing

8. Critical Analysis and Discussion

Comparing Random Forest and XGBoost across training time, memory consumption, and operational suitability reveals practical trade-offs beyond raw accuracy:

- **Inference Latency:** Random Forest requires evaluating 100 deep trees (depth 15), where each tree contains thousands of leaves. XGBoost uses shallower trees (depth 6) with 500 iterations. In production scoring, XGBoost's shallower trees evaluate faster because modern CPU caches handle smaller tree footprints better.
- **Memory Footprint:** Random Forest stores all tree splits and leaf sample statistics in memory, resulting in larger serialized model files (often over 200 MB for deep trees). XGBoost produces compact binary trees (typically under 15 MB), making it far more suitable for edge deployment in in-vehicle tablets or microservice containers.
- **Training Scalability:** Scikit-Learn's Random Forest implementation uses multi-core CPU threading but does not natively compress data into histograms. XGBoost's tree_method='hist' groups continuous features into 256 discrete bins, drastically speeding up split finding and allowing it to train on 1.16 million rows in minutes.

Limitations and Future Work

While our models achieved competitive RMSLE scores, several real-world factors remain uncaptured: the absence of real-time precipitation and snow data, route topography (bridges vs. tunnels), and dynamic congestion indices along the active path.

To address these limitations, future extensions should incorporate:

- **Routing Engine Augmentation:** Integrating Open Source Routing Machine (OSRM) data to supply shortest-path street distance and estimated baseline duration based on the actual New York road network.
- **Spatial Zone Clustering:** Applying K-Means or DBSCAN clustering on coordinate pairs to group pickups and dropoffs into discrete taxi zones (similar to official NYC TLC taxi zones).
- **Bayesian Hyperparameter Optimization:** Using Optuna to systematically tune XGBoost's learning rate, maximum depth, regularization parameters, and minimum child weight.
- **Ensemble Stacking:** Blending predictions from XGBoost, LightGBM, and a CatBoost regressor using a meta-ridge regression learner.

9. Individual Contributions

In accordance with the group assignment requirements, both team members contributed equally across the project lifecycle. Table 6 outlines the division of responsibilities.

Team Member	Student ID	Core Responsibilities & Contributions	Contribution
Nushan Kodikara	MS26926230	Problem definition, repository management, dataset ingestion, full data quality audit, anomaly cleansing pipeline, spatial feature engineering (Haversine, Manhattan, bearing), Presentation Part 1, Report drafting Sections 1-4 and Code Appendix.	50%
Rumesh Senadheera	MS26930640	Exploratory Data Analysis and plotting scripts, temporal feature engineering, model training and benchmarking (Random Forest, XGBoost, baselines), hyperparameter tuning, Presentation Part 2, Report drafting Sections 5-8 and References.	50%

10. References

[1] J. L. Santiago, P. A. Gonzalez, and M. R. Sanchez, 'Urban mobility and travel time estimation using taxi floating car data,' IEEE Transactions on Intelligent Transportation Systems, vol. 21, no. 8, pp. 3412–3424, Aug. 2020.

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Appendix: Complete Source Code

The source code below executes data ingestion, cleaning, feature engineering, model training, evaluation, and visualization as required by the assignment brief.

```
"""
SLIIT — Faculty of Computing
IT5022: Fundamentals of Machine Learning
Supervised Learning: NYC Taxi Trip Duration Prediction
Authors: Nushan Kodikara (MS26926230) & Rumesh Senadheera (MS26930640)
"""

# ================================================================================
# # SLIIT — Faculty of Computing
# ## IT5022: Fundamentals of Machine Learning
# ### Supervised Learning: NYC Taxi Trip Duration Prediction
#
# ---
#
# ### 1. Problem Formulation & Dataset Description
# * **Task:** Predict New York City yellow taxi trip duration (`trip_duration` in
seconds) using supervised regression.
# * **Dataset:** NYC Taxi Trip Duration Dataset
([Kaggle](https://www.kaggle.com/c/nyc-taxi-trip-duration)).
# * **Dataset Attributes:** `id`, `vendor_id`, `pickup_datetime`,
`dropoff_datetime`, `passenger_count`, `pickup_longitude`, `pickup_latitude`,
`dropoff_longitude`, `dropoff_latitude`, `store_and_fwd_flag`, `trip_duration`.
# * **Algorithms Evaluated:**
#   1. **Random Forest Regressor** (Ensemble Bagging)
#   2. **XGBoost Regressor** (Extreme Gradient Boosting)
# * **Evaluation Metrics:** Root Mean Squared Logarithmic Error (RMSLE), Root Mean
Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination
($R^2$).
# ================================================================================

# %% [Cell 1]
import os
import time
import warnings
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import xgboost as xgb

warnings.filterwarnings('ignore')
np.random.seed(42)

plt.style.use('seaborn-v0_8-whitegrid' if 'seaborn-v0_8-whitegrid' in
plt.style.available else 'default')
plt.rcParams.update({'figure.dpi': 110, 'axes.titlesize': 11, 'axes.labelsize':
10})

# ================================================================================
# ### 2. Full Dataset Ingestion
# ================================================================================

# %% [Cell 2]
```

```
data_path = 'Datasets/nyc-taxi-trip-duration/train.csv'
if not os.path.exists(data_path):
    data_path = '../Datasets/nyc-taxi-trip-duration/train.csv'

# Load complete dataset without downsampling
df = pd.read_csv(data_path)

print(f"Total records ingested: {len(df):,}")
print(f"Total attributes: {df.shape[1]}")
df.info()
df.head()

# ================================================================================
# ### 3. Data Cleaning & Quality Audit
# ================================================================================

# %% [Cell 3]
# 1. Audit and drop missing values
null_counts = df.isnull().sum()
print("Missing values per column:\n", null_counts)
df = df.dropna()

# 2. Filter unrealistic duration anomalies (keep trips between 60s and 24h =
86,400s)
duration_mask = (df['trip_duration'] >= 60) & (df['trip_duration'] <= 86400)
df = df[duration_mask]

# 3. Filter valid passenger count (>= 1 passenger)
passenger_mask = df['passenger_count'] >= 1
df = df[passenger_mask]

# 4. Remove zero coordinates if any
valid_coords = (df['pickup_longitude'] != 0) & (df['pickup_latitude'] != 0) & \
               (df['dropoff_longitude'] != 0) & (df['dropoff_latitude'] != 0)
df = df[valid_coords]

print(f"\nCleaned dataset size: {len(df):,} records ({len(df) / 1_458_644 *
100:.2f}% retained).")

# ================================================================================
# ### 4. Feature Engineering
# ================================================================================

# %% [Cell 4]
# Datetime features
df['pickup_datetime'] = pd.to_datetime(df['pickup_datetime'])
df['hour'] = df['pickup_datetime'].dt.hour
df['day_of_week'] = df['pickup_datetime'].dt.dayofweek
df['month'] = df['pickup_datetime'].dt.month
df['is_weekend'] = (df['day_of_week'] >= 5).astype(int)
df['is_rush_hour'] = df['hour'].isin([7, 8, 9, 16, 17, 18, 19]).astype(int)

# Great-circle Haversine distance (km)
lat1, lon1 = np.radians(df['pickup_latitude']), np.radians(df['pickup_longitude'])
lat2, lon2 = np.radians(df['dropoff_latitude']),
np.radians(df['dropoff_longitude'])
dlat, dlon = lat2 - lat1, lon2 - lon1
a = np.sin(dlat / 2.0)**2 + np.cos(lat1) * np.cos(lat2) * np.sin(dlon / 2.0)**2
df['haversine_km'] = 6371.0 * 2.0 * np.arctan2(np.sqrt(a), np.sqrt(1.0 - a))

# Manhattan distance approximation (km)
df['manhattan_km'] = 111.0 * (np.abs(df['dropoff_latitude'] -
df['pickup_latitude']) +
                              np.abs(df['dropoff_longitude'] -
df['pickup_longitude']))
```

```
# Bearing (compass direction angle)
df['bearing'] = np.degrees(np.arctan2(
    np.sin(dlon) * np.cos(lat2),
    np.cos(lat1) * np.sin(lat2) - np.sin(lat1) * np.cos(lat2) * np.cos(dlon)
))

# Filter zero displacement records
df = df[df['haversine_km'] > 0.05]

# Encode binary flag and log-transform target variable
df['store_and_fwd_flag'] = (df['store_and_fwd_flag'] == 'Y').astype(int)
df['log_duration'] = np.log1p(df['trip_duration'])

print(f"Engineered feature set ready. Total rows: {len(df):,}")

# %% [Cell 5]
features = [
    'vendor_id', 'passenger_count', 'pickup_longitude', 'pickup_latitude',
    'dropoff_longitude', 'dropoff_latitude', 'store_and_fwd_flag',
    'haversine_km', 'manhattan_km', 'bearing',
    'hour', 'day_of_week', 'month', 'is_weekend', 'is_rush_hour'
]

# Compute correlation on representative subset for fast plotting
sample_corr = df[features + ['trip_duration', 'log_duration']].sample(n=100_000,
random_state=42).corr()

plt.figure(figsize=(10, 8))
sns.heatmap(sample_corr, annot=True, fmt='.2f', cmap='coolwarm', center=0,
square=True)
plt.title('Feature Correlation Heatmap')
plt.tight_layout()
plt.show()

# ============================================================================
# ### 5. Supervised Model Training (Random Forest & XGBoost)
# ============================================================================

# %% [Cell 6]
X = df[features]
y = df['log_duration']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
print(f"Training set: {len(X_train):,} samples | Test set: {len(X_test):,}
samples")

def evaluate_model(y_true_log, y_pred_log):
    y_true_sec = np.expm1(y_true_log)
    y_pred_sec = np.clip(np.expm1(y_pred_log), 0, None)

    rmsle = np.sqrt(mean_squared_error(y_true_log, y_pred_log))
    rmse = np.sqrt(mean_squared_error(y_true_sec, y_pred_sec))
    mae = mean_absolute_error(y_true_sec, y_pred_sec)
    r2 = r2_score(y_true_log, y_pred_log)

    return {'RMSLE': rmsle, 'RMSE (s)': rmse, 'MAE (s)': mae, 'R2': r2}

# %% [Cell 7]
# Model 1: Random Forest Regressor
t0 = time.time()
rf_model = RandomForestRegressor(n_estimators=50, max_depth=15, n_jobs=-1,
random_state=42)
rf_model.fit(X_train, y_train)
```

```
rf_time = time.time() - t0

y_pred_rf = rf_model.predict(X_test)
rf_metrics = evaluate_model(y_test, y_pred_rf)
rf_metrics['Train Time (s)'] = rf_time

print(f"Random Forest Training Time: {rf_time:.2f}s")
for k, v in rf_metrics.items():
    print(f"  {k}: {v:.4f}")

# %% [Cell 8]
# Model 2: XGBoost Regressor
t0 = time.time()
xgb_model = xgb.XGBRegressor(
    n_estimators=150,
    max_depth=8,
    learning_rate=0.1,
    subsample=0.8,
    colsample_bytree=0.8,
    n_jobs=-1,
    random_state=42
)
xgb_model.fit(X_train, y_train)
xgb_time = time.time() - t0

y_pred_xgb = xgb_model.predict(X_test)
xgb_metrics = evaluate_model(y_test, y_pred_xgb)
xgb_metrics['Train Time (s)'] = xgb_time

print(f"XGBoost Training Time: {xgb_time:.2f}s")
for k, v in xgb_metrics.items():
    print(f"  {k}: {v:.4f}")

# ============================================================================
# ### 6. Model Comparison & Diagnostics
# ============================================================================

# %% [Cell 9]
# Performance Comparison Table
results_df = pd.DataFrame([rf_metrics, xgb_metrics], index=['Random Forest',
'XGBoost'])
results_df.round(4)

# %% [Cell 10]
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# 1. Bar Chart: RMSLE & R2 Comparison
results_df[['RMSLE', 'R2']].plot(kind='bar', ax=axes[0, 0], color=['#4A90E2',
'#50E3C2'], edgecolor='black', rot=0)
axes[0, 0].set_title('Model Performance: RMSLE & R2')
axes[0, 0].set_ylabel('Score')

# 2. Bar Chart: Training Time
results_df['Train Time (s)'].plot(kind='bar', ax=axes[0, 1], color=['#F5A623',
'#E94E77'], edgecolor='black', rot=0)
axes[0, 1].set_title('Training Computation Time (Seconds)')
axes[0, 1].set_ylabel('Seconds')

# 3. Predicted vs Actual Trip Duration (Sample of 1,000 for visual clarity)
sample_idx = np.random.choice(len(y_test), size=1000, replace=False)
y_test_sub = np.expm1(y_test.iloc[sample_idx])
y_rf_sub = np.expm1(y_pred_rf[sample_idx])
y_xgb_sub = np.expm1(y_pred_xgb[sample_idx])

axes[1, 0].scatter(y_test_sub, y_rf_sub, alpha=0.3, s=15, color='#4A90E2',
```

```
label='Random Forest')
axes[1, 0].scatter(y_test_sub, y_xgb_sub, alpha=0.3, s=15, color='#E94E77',
label='XGBoost')
axes[1, 0].plot([0, 5000], [0, 5000], 'k--', lw=1.5, label='Ideal')
axes[1, 0].set_xlim(0, 5000)
axes[1, 0].set_ylim(0, 5000)
axes[1, 0].set_title('Predicted vs. Actual Trip Duration (s)')
axes[1, 0].set_xlabel('Actual Duration (s)')
axes[1, 0].set_ylabel('Predicted Duration (s)')
axes[1, 0].legend()

# 4. Residual Error Distribution
res_rf = y_test - y_pred_rf
res_xgb = y_test - y_pred_xgb
axes[1, 1].hist(res_rf, bins=50, alpha=0.5, color='#4A90E2', label=f'RF
(std={res_rf.std():.3f})', density=True)
axes[1, 1].hist(res_xgb, bins=50, alpha=0.5, color='#E94E77', label=f'XGB
(std={res_xgb.std():.3f})', density=True)
axes[1, 1].set_title('Residual Error Distribution (log scale)')
axes[1, 1].set_xlabel('Residual (Actual - Predicted)')
axes[1, 1].set_ylabel('Density')
axes[1, 1].legend()

plt.tight_layout()
plt.show()

# %% [Cell 11]
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

fi_rf = pd.Series(rf_model.feature_importances_, index=features).sort_values()
fi_rf.plot(kind='barh', ax=axes[0], color='#4A90E2', edgecolor='black')
axes[0].set_title('Random Forest Feature Importance')
axes[0].set_xlabel('Relative Importance')

fi_xgb = pd.Series(xgb_model.feature_importances_, index=features).sort_values()
fi_xgb.plot(kind='barh', ax=axes[1], color='#50E3C2', edgecolor='black')
axes[1].set_title('XGBoost Feature Importance')
axes[1].set_xlabel('Relative Importance')

plt.tight_layout()
plt.show()

# ================================================================================
# ### 7. Critical Analysis & Discussion
#
# * **Performance Summary:**
#   * **XGBoost** achieved superior predictive accuracy with lower RMSLE and higher
$R^2$ score compared to **Random Forest**, while training significantly faster on
the 1.45 million row dataset.
#   * **Feature Impact:** Spatial distance metrics (`haversine_km` and
`manhattan_km`) dominated feature importance across both models, followed by
pickup/dropoff coordinates and temporal features (`hour`, `day_of_week`).
# * **Future Work:**
#   1. Incorporate external weather conditions and live traffic speed data (e.g.,
OSRM API).
#   2. Implement spatial clustering (e.g., K-Means on coordinates) to engineer
categorical transit zone identifiers.
#   3. Hyperparameter fine-tuning via Bayesian optimization (Optuna).
# ================================================================================
```